E0 297: Cryptographic Hardware Accelerators

Paper Review Presentation Script

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Reviewer: Anirudh **Track:** Track 2 (SHA2 / SHA3 Accelerators)
Paper: *18Gbps, 50mW Reconfigurable Multi-Mode SHA Hashing Accelerator in 45nm CMOS*
Authors & Venue: R. Ramanarayanan et al., *IEEE ISSCC 2010* **Duration:** ∼3.5–4 Minutes

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1. Introduction & Problem Formulation *∼45 seconds*

"Good morning / afternoon, sir and everyone.

Today I am presenting my review of the paper **'18Gbps, 50mW Reconfigurable Multi-Mode SHA Hashing Accelerator in 45nm CMOS'**, authored by Ramanarayanan et al. from Intel Labs and presented at ISSCC 2010.

The fundamental design challenge this work addresses is **hardware reconfigurability for secure hashing**. Security protocols like IPsec and TLS require support across multiple standards: SHA-1, SHA-224, SHA-256, SHA-384, and SHA-512. Traditionally, hardware designers either implement separate, dedicated datapaths for each algorithm—which incurs huge silicon area overhead—or resort to microcoded firmware, which lacks line-rate throughput.

The authors overcome this tradeoff by proposing a **single, unified 64-bit bitslice datapath** in 45nm CMOS that supports all five SHA variants using just three configuration bits ($m[2:0]$). Fabricated on silicon, the chip hits **1.4 GHz** at 1.1 V, delivering **18 to 23 Gbps** of throughput at only **50 mW**, achieving approximately **40% area reduction** compared to dedicated engines with under 3% reconfiguration delay penalty."

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2. Key Architectural Strengths *∼1.5 minutes*

"When evaluating the design against our course concepts, three architectural strengths stand out:

1. Unified Bit-Slice and Cross-Algorithm Register Reuse:
Both SHA-1 and SHA-2 belong to the Merkle–Damgård family, relying on round-based iterative compression. However, SHA-1 operates on a 5-word 32-bit state, whereas SHA-2 operates on an 8-word state of either 32-bit or 64-bit width. The authors observed that SHA-1's critical multi-operand update for register A mirrors the 6-operand addition tree of SHA-2's register E.

Consequently, they mapped SHA-1 registers $\{A, B, C, D, E\}$ directly onto SHA-2 circuits $\{E, D, F, G, H\}$, gating off SHA-2 circuits A, B, and C. Furthermore, SHA-1's 30-bit left rotate is mapped natively as a 2-bit right rotate ($\text{ROTL}^{30} \equiv \text{ROTR}^{2}$), eliminating dedicated rotation logic. This allows near-total adder reuse with negligible muxing delay.

2. Four-Stage Carry-Save Compressor Tree:
In SHA-2, updating state register A requires summing 8 operands in a single cycle: $K_t, W_t, h, Ch, \Sigma_1, d, \Sigma_0$, and Maj. Cascading standard carry-propagate adders sequentially would severely limit the clock frequency. The authors maintain intermediate sums in redundant carry-save representation through **four stages of 3:2 compressors**. Carry propagation is deferred to the very end of the cycle, executed once by a dual 64-bit adder pair for A and E, enabling a 1.4 GHz clock.

3. Bit-32 Carry-Kill and Dual-Stream Parallelism:
In 32-bit modes (SHA-1/224/256), a 64-bit datapath would ordinarily suffer 50% underutilization. By inserting a **carry-kill gate at bit 32**, the 64-bit datapath splits cleanly into two independent 32-bit lanes. This enables processing two independent hash blocks concurrently, doubling throughput to 23 Gbps without duplicate compute resources, while clock-gating unused register file partitions saves dynamic power."

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3. Critical Shortcomings & Weaknesses *~1 minute*

"However, looking at the design critically from an end-to-end accelerator perspective:

1. The Message Expansion Scheduler is Excluded:
The reported area (0.0625 mm^2) and power (50 mW) cover only the compression core. The message scheduler, which expands 16 message words into 64 or 80 schedule words using shifts, XORs, and modular adders, is left out. In real-world systems and Bitcoin mining ASICs where energy-per-hash across the entire chip is the governing metric, omitting the scheduler paints an incomplete picture of system efficiency.

2. Carry-Propagate Adders Bottleneck Low-Voltage Scaling:
Even with carry-save trees, the two 64-bit carry-propagate adders at the output stage remain on the critical path. As supply voltage scales into the near-threshold regime for ultra-low-power IoT applications, carry-chain delay degrades rapidly. The authors' own measurements show $F_{\max}$ collapsing from 1.4 GHz down to **21 MHz at 320 mV**, demonstrating that this design cannot operate efficiently near threshold.

3. Incompatibility with Modern Sponge Ciphers:
The core is specialized for Merkle–Damgård modular arithmetic. It cannot support modern lightweight or post-quantum hash standards like SHA-3 (Keccak) or ASCON, which rely on wide-state (1600-bit or 320-bit) bitwise permutations without modular additions. Although reasonable for a 2010 publication, the hardware cannot adapt to modern sponge cryptography."

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4. Conclusion & Recommendation *~30 seconds*

"To conclude:

- **Reviewer Confidence: Medium.** I fully comprehend the datapath mapping, compressor sizing, and architectural tradeoffs, though I have not replicated the 45nm layout simulations.
- **Overall Rating: Good.** The paper demonstrates genuine silicon innovation in unified bitslice design and high-frequency CSA trees, balanced by practical omissions in message scheduling and near-threshold scaling.

Thank you. I am happy to take any questions or discuss further."

Quick Q&A Defense Sheet

- **Q: Why map SHA-1's A to SHA-2's E instead of A?**
  Defense: SHA-1 register A needs a 5-operand modular sum ($A + f + E + K_t + W_t$). SHA-2 register E has a matching multi-operand adder tree ($d + h + \Sigma_1 + Ch + K_t + W_t$). Mapping SHA-1 $A \to E$ reuses the existing compressor hardware rather than creating new paths.

- **Q: Why does the paper count 8 operands in SHA-2?**
  Defense: In SHA-2, $A_{new} = T_1 + T_2$ where $T_1 = h + \Sigma_1(e) + Ch(e, f, g) + K_t + W_t$ (5 operands) and $T_2 = \Sigma_0(a) + Maj(a, b, c)$ (2 operands). Together with forwarding d for register E, the tree condenses 8 operands in total before the final CPAs.

- **Q: What is the left-rotate conversion trick?**
  Defense: In a 32-bit register, a 30-bit circular left shift is identical to a 2-bit circular right shift ($\text{ROTL}^{30} \equiv \text{ROTR}^{32-30} = \text{ROTR}^{2}$). Because SHA-2 already implements right-rotators (Σ_0, Σ_1), the datapath reuses right-rotation routing directly.